Year 13 Maths - Applications of vectors

Similar questions - Constant acceleration in vector form

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A particle P moves with constant acceleration $(2\mathbf{i} - 3\mathbf{j})\ \mathrm{m\,s^{-2}}$. At time $t = 5$ seconds, the velocity of P is $(4\mathbf{i} + \mathbf{j})\ \mathrm{m\,s^{-1}}$.

(a) Determine the speed of P when $t = 0$. **[4]**

(b) Find the displacement of P between $t = 0$ and $t = 4$. **[3]**

2 A particle Q is moving with constant acceleration $(-3\mathbf{i} + 4\mathbf{j})\ \mathrm{m\,s^{-2}}$. At time $t = 0$, Q is at the origin with velocity $(2\mathbf{i} - \mathbf{j})\ \mathrm{m\,s^{-1}}$.

(a) Find, in terms of $\mathbf{i}$ and $\mathbf{j}$, the displacement of Q at $t = 3$. **[2]**

(b) Find the speed of Q at $t = 3$. **[3]**

(c) Find the time at which Q is moving parallel to the vector $\mathbf{j}$ (that is, due north). **[3]**

Question 1

(a) $\mathbf{u} = \mathbf{v} - \mathbf{a}t = (4\mathbf{i}+\mathbf{j}) - 5(2\mathbf{i}-3\mathbf{j}) = -6\mathbf{i}+16\mathbf{j}$. Speed $= \sqrt{36+256} = \sqrt{292} = 17.1\ \text{m s}^{-1}$

(b) $\mathbf{s} = \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2 = 4(-6\mathbf{i}+16\mathbf{j}) + 8(2\mathbf{i}-3\mathbf{j}) = -8\mathbf{i}+40\mathbf{j}$ m

Question 2

(a) $\mathbf{s} = 3(2\mathbf{i}-\mathbf{j}) + \frac{9}{2}(-3\mathbf{i}+4\mathbf{j}) = -7.5\mathbf{i}+15\mathbf{j}$

(b) $\mathbf{v} = (2\mathbf{i}-\mathbf{j}) + 3(-3\mathbf{i}+4\mathbf{j}) = -7\mathbf{i}+11\mathbf{j}$, speed $= \sqrt{49+121} = \sqrt{170} = 13.0\ \text{m s}^{-1}$

(c) The $\mathbf{i}$-component of velocity is $2-3t = 0$, so $t = \frac{2}{3}$ s